Frequency-tagged visual paradigms for short, high-signal EEG: SSVEP, FPVS face individuation, and the Fastball memory assessment

Morgan Hough

`eegnb.reports.build_review`

April 18, 2026

Abstract

Three closely related EEG paradigms share a common logic: stimulate the visual system at a precisely fixed periodic rate, and read out the cortical response in the frequency domain rather than by trial-averaging in the time domain. Classical SSVEP, reviewed definitively by Norcia et al. (2015), establishes the frequency-tagging principle and its signal-to-noise advantages. The Rossion laboratory's fast periodic visual stimulation oddball (Liu-Shuang et al., 2014) adapts the principle to probe category-level discrimination — most notably face individuation. Stothart et al. (2020, 2021, 2025) build on the same machinery to produce "Fastball", a clinically-oriented recognition-memory assay that takes under five minutes of EEG per subject and is already being evaluated as a pre-dementia biomarker. This document reviews the three paradigms, their shared methodology, their divergent applications, and their practical implications for low-cost consumer-grade EEG such as the g.tec Unicorn Hybrid Black used in this project.

Contents

1 Introduction

1.1 The signal-to-noise problem in EEG

Scalp EEG recovers signals of the order of 1–50 µV from an envelope of 10–100 µV of background activity and movement artifact. The dominant strategy for isolating a cognitive signal is time-domain averaging: present a stimulus many times, epoch the EEG, and average the epochs. Signal-to-noise ratio (SNR) then grows as $\sqrt{N}$ where N is the trial count, producing the familiar event-related potentials — N170, P300, N400 — each of which typically requires 100 or more trials and 15 to 25 minutes of recording per condition.

1.2 The frequency-tagging alternative

An alternative, developed over decades in vision science, exploits the fact that a sinusoidally or periodically modulated stimulus evokes a response at the same frequency. If stimulation is at f Hz, the cortical response concentrates at f Hz and its harmonics; noise in the EEG is broadband ($\sim 1/f$ plus measurement). A narrow-band SNR estimate therefore yields a dramatic signal-to-noise gain without trial averaging. Norcia et al. (2015) review the full SSVEP literature and codify the analysis principles.

The insight that matters here is that *two* frequencies can be tagged simultaneously. If standard stimuli appear at a base frequency F and oddball stimuli appear at F/n, then a cortical response at F/n and its harmonics specifically indicates discrimination between standard and oddball — without any behavioural task. This is the core of both the Rossion face-individuation paradigm and the Stothart Fastball memory assessment.

2 Steady-state visual evoked potentials (SSVEP)

2.1 Definition and taxonomy

The SSVEP is the cortical response to periodic visual stimulation, stationary in the frequency domain. Norcia et al. (2015) distinguish:

- **On–off flicker** at a fixed frequency (most common in BCI and attention work).
- **Continuous sinusoidal contrast modulation**, preferred in Rossion's face-individuation paradigm for its gentler spectral leakage.
- **Multi-frequency tagging**, in which different stimulus elements flicker at different rates to permit simultaneous tracking of attentional state across competing streams.

2.2 Why SSVEP beats trial-averaged ERP on SNR

Let $x(t)$ be EEG recorded during steady-state stimulation with period $T = 1/f$. The signal is present at f and its harmonics $\{2f, 3f, \ldots\}$ as a line spectrum; the background EEG is approximately $1/f$-distributed with broadband noise overlaid. A narrow-band SNR estimate

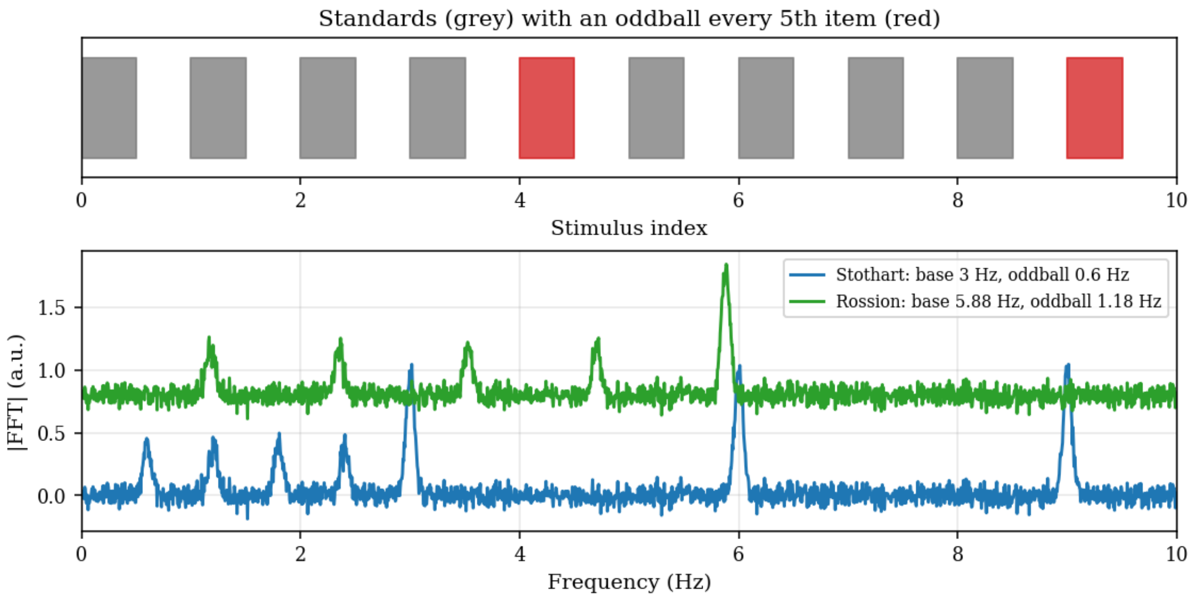

Figure 1: Schematic timing of a frequency-tagged oddball paradigm. Every 5th standard is replaced by an oddball, yielding responses at the base rate F and at the oddball rate $F/5$. Top bar: stimulus stream. Bottom traces: idealised amplitude spectra as base rate rises from 3 Hz (Stothart Fastball) to 5.88 Hz (Rossion FPVS).

$$\mathrm{SNR}(f) = \frac{|X(f)|}{\frac{1}{|B|}\sum_{f' \in B} |X(f')|} \tag{1}$$

with B a set of neighbouring non-target bins, grows with recording duration approximately as $\sqrt{T}$ (because the DFT bin width narrows as $1/T$), and the line component grows linearly in amplitude. This is why 60–180 s of SSVEP data can match or exceed the per-subject SNR of a 20 min averaged-ERP study.

2.3 Typical parameters

- Base frequencies in the literature span 2–40 Hz; face-specific work uses 5–6 Hz (see Section 3).
- Session durations of 60–180 s per condition are standard.
- Electrodes: medial occipital (Oz) for generic SSVEP; lateral occipito-temporal (PO7/PO8/P9/P10) for face-specific responses.

3 FPVS face individuation (Rossion lab)

3.1 Paradigm

Liu-Shuang et al. (2014) introduced an oddball variant of FPVS for unfamiliar face individuation. An image of a single unfamiliar base identity is presented at 5.88 Hz (170 ms SOA); every fifth stimulus is swapped for a different unfamiliar identity, yielding an oddball rate of 1.176 Hz. A response at the oddball frequency and its harmonics indicates that the visual system discriminates the newly-presented identity from the repeated base.

3.2 Key empirical results

- A robust oddball response is recorded over *right* occipito-temporal cortex (PO8, P10), convergent with the known right-hemisphere dominance of face identity processing.
- Intracerebral recordings localise the response to inferior occipital and fusiform gyri.
- The paradigm has now been applied in more than twenty published studies covering prosopagnosia, autism spectrum conditions, infant face processing, and cross-cultural face recognition.

3.3 Stimuli

The canonical Liu-Shuang stimuli are 50 unfamiliar face identities, neutral expression, cropped to exclude external features, presented against a grey background at $11° \times 9°$ of visual angle. These stimuli are not publicly distributed due to subject-image rights. Open alternatives with comparable norming include the Chicago Face Database and KDEF; custom sets compiled from either can reproduce the paradigm faithfully with minor re-norming.

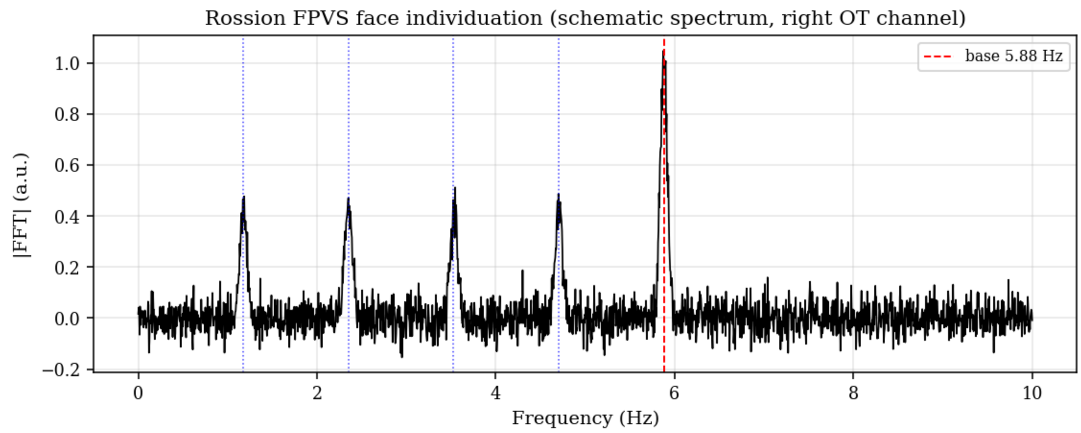

Figure 2: Idealised Rossion FPVS oddball response at right occipito-temporal cortex. Red dashed line: base 5.88 Hz. Blue dashed lines: oddball 1.176 Hz and harmonics. The oddball response is the unambiguous neural signature of face individuation.

4 Fastball memory assessment (Stothart et al.)

4.1 Paradigm

Stothart et al. (2020) introduced the Fastball paradigm as a recognition-memory variant of frequency-tagged oddball. Every-day colour objects drawn from the Bank of Standardised Stimuli v2.0 (Brodeur et al., 2010) are shown at a base rate of 3 Hz (166 ms image + 166 ms ISI). In the recognition condition, every fifth image is an item that the participant has been pre-exposed to moments earlier; the oddball rate is 0.6 Hz. An above-baseline response at 0.6 Hz and harmonics demonstrates that the cortex differentiates seen from unseen items, *with no behavioural response required.*

4.2 Application to dementia

- Stothart et al. (2021) applied Fastball to twenty Alzheimer's patients, twenty healthy older adults, and twenty younger adults. The paradigm distinguished AD from healthy older adults with AUC $= 0.86$ ($p < 0.001$), whereas a behavioural two-alternative forced-choice task achieved only AUC $= 0.63$ ($p = 0.148$) on the same cohort.

- Stothart et al. (2025) extended the paradigm to 107 participants including amnestic and non-amnestic mild cognitive impairment. Fastball responses in aMCI were reduced relative to controls, supporting use as an early biomarker of amnestic dysfunction.
- Home-use trials are ongoing in collaboration with BRACE Alzheimer's Research; results are forthcoming in 2026.

4.3 Why it works for memory specifically

The Fastball oddball is a *memory-modulated* stimulus: whether a given image counts as "standard" or "oddball" is determined by the participant's history, not by any low-level visual feature. Discriminability in the frequency-domain response therefore cannot be inherited from basic sensory mismatch — it requires access to encoding and retrieval circuitry. In Alzheimer's disease the retrieval disruption is profound enough that the oddball response collapses even when patients still score above chance on overt tests, which is precisely what makes the paradigm a useful early biomarker.

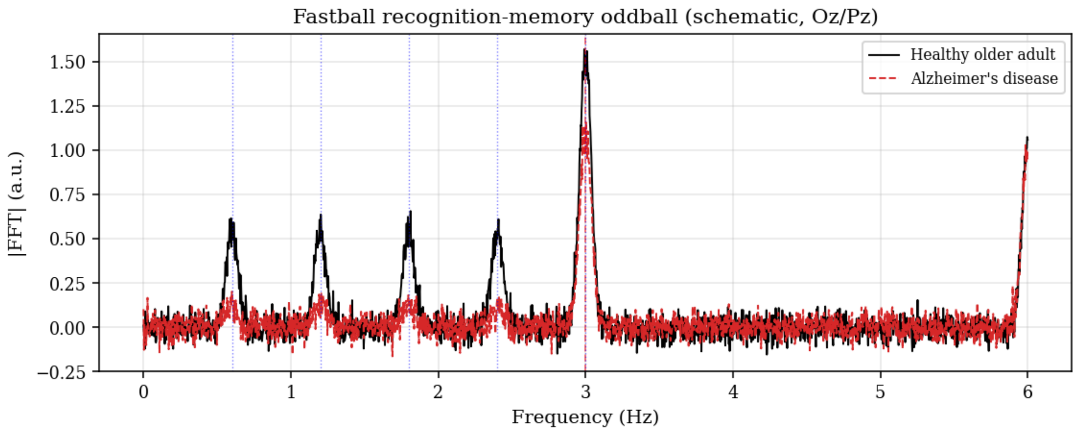

Figure 3: Idealised Stothart Fastball spectrum at Oz/Pz. Red dashed line: base 3 Hz. Blue dashed lines: oddball 0.6 Hz and harmonics. In Alzheimer's disease (dashed trace) the oddball response amplitude is markedly reduced relative to healthy older adults (solid trace), while the base-rate response is preserved.

5 Shared methodology: analysis of a frequency-tagged oddball

Across all three paradigms, the analysis pipeline is largely shared.

5.1 Pre-processing

1. Notch filter at line frequency (50 Hz or 60 Hz).
2. Bandpass 0.1 to 100 Hz (broad enough to retain harmonics).
3. Artifact rejection by amplitude threshold or component-based methods (ICA for multi-channel montages; simple threshold for 8-channel dry-electrode devices such as the g.tec Unicorn).
4. Crop to integer multiples of the oddball period so the FFT has bins exactly at the target frequencies (avoids spectral leakage).

5.2 Spectral analysis

Compute the FFT of the continuous pre-processed record per channel. Extract amplitudes at the target (base and oddball) frequencies and their harmonics.

5.3 Signal-to-noise estimation

The Liu-Shuang / Stothart convention is

$$\mathrm{SNR}(f_t) \;=\; \frac{|X(f_t)|}{\frac{1}{|N|}\sum_{f\in N}|X(f)|} \tag{2}$$

where N is the set of neighbouring bins within ± 0.10 Hz of the target, excluding the immediately-adjacent bins. A z-score of the target bin against the noise distribution, $z > 1.96$, corresponds to $p < 0.05$ uncorrected and is the standard per-harmonic significance threshold.

5.4 Harmonic summation

Summing SNR across harmonics $\{f_t, 2f_t, 3f_t, \ldots\}$ increases sensitivity when the response contains energy beyond the fundamental — which is the norm in visual oddball paradigms. Stothart sums ten harmonics up to 7.2 Hz; Rossion sums four to five harmonics up to 5 Hz.

6 Practical implications for low-channel EEG

The g.tec Unicorn Hybrid Black provides eight fixed dry electrodes (Fz, C3, Cz, C4, Pz, PO7, Oz, PO8) at 250 Hz. This is materially sparser than the 65–128 channel research montages in the primary papers, but it is sufficient for:

- **SSVEP generic**: Oz and the parieto-occipital ring (PO7/PO8) are the primary SSVEP sites — the Unicorn covers all three.
- **Rossion face FPVS**: PO8 proxies the right occipito-temporal peak reported by Liu-Shuang. PO7 provides the left-hemisphere control. Oz captures the base-rate response. Missing: P10 / TP10, which sit ventrally relative to PO8 and capture additional face-response variance but are not essential.
- **Stothart Fastball**: Oz and Pz cover the mid-occipital/ parietal ridge where Stothart reports peak oddball amplitude.

In this project, the new paradigm modules (`eegnb.experiments.fpvs_stothart.VisualFPVSStothart` and `eegnb.experiments.fpvs_rossion.VisualFPVSRossion`) expose the paradigm-appropriate region of interest as a `channels_of_interest` attribute, which downstream analysis code resolves to brainflow channel indices via `eegnb.utils.rois.resolve_roi`. Because the Unicorn's hardware montage is fixed, this amounts to a software-side ROI selection rather than a physical reconfiguration.

7 Limitations and open questions

- **Display timing.** Frequency-tagged paradigms require precise per-frame timing; lab-grade CRTs or gaming-grade LCDs at refresh rates that are integer multiples of the target frequency are preferred. On Wayland-native compositors (Fedora 43 etc.) XWayland introduces frame jitter of the order of the monitor's vblank interval; whether this is tolerable depends on the target precision.

- **Dry electrodes.** Stothart's validation was run on a 65-channel HydroCel Geodesic net with impedances <50 kΩ. Dry-electrode devices such as the Unicorn start an order of magnitude higher and are more sensitive to motion artifact. Expect to need more trials/longer sessions than the published minimums.
- **Reference identities.** Rossion's paradigm as published fixes the base identity throughout a sequence. Replacing the base with a pool of identities changes the inference: the oddball response then indexes discrimination between two pools, not between a specific identity and the rest.
- **Clinical translation.** Stothart's Alzheimer's result is impressive at AUC = 0.86 but is based on a relatively small sample; community-wide validation and home-use data are still emerging.

8 Conclusion

Three paradigms, one trick: periodic stimulation turns a trial-averaging problem into a narrow-band SNR problem. SSVEP demonstrates the principle, Rossion FPVS adapts it to category-level discrimination with a face-identity oddball, and Stothart's Fastball extends it into a clinically viable memory assay. For small-channel dry-electrode EEG systems such as the g.tec Unicorn, all three remain tractable with minor ROI accommodations. The shared logic means the underlying analysis pipeline — FFT, SNR at target plus harmonics, z-score against neighbouring bins — is portable across paradigms with minimal code change.

References

Norcia, A. M., Appelbaum, L. G., Ales, J. M., Cottereau, B. R., and Rossion, B. (2015). The steady-state visual evoked potential in vision research: A review. *Journal of Vision* 15(6):4. doi:10.1167/15.6.4.

Liu-Shuang, J., Norcia, A. M., and Rossion, B. (2014). An objective index of individual face discrimination in the right occipito-temporal cortex by means of fast periodic oddball stimulation. *Neuropsychologia* 52:57–72. doi:10.1016/j.neuropsychologia.2013.10.022.

Stothart, G., Smith, L. J., Milton, A., and Coulthard, E. (2020). A rapid, neural measure of implicit recognition memory using fast periodic visual stimulation. *NeuroImage* (paradigm introduction paper; DOI to confirm).

Stothart, G., Smith, L. J., Milton, A., and Coulthard, E. (2021). A passive and objective measure of recognition memory in Alzheimer's disease using Fastball memory assessment. *Brain* 144(9):2812–2825. doi:10.1093/brain/awab154.

Stothart, G. et al. (2025). A passive and objective measure of recognition memory in mild cognitive impairment using Fastball memory assessment. *Brain Communications* 7(5):fcaf279. doi:10.1093/braincomms/fcaf279.

Brodeur, M. B., Dionne-Dostie, E., Montreuil, T., and Lepage, M. (2010). The Bank of Standardized Stimuli (BOSS), a new set of 480 normative photos of objects to be used as visual stimuli in cognitive research. *PLoS ONE* 5(5):e10773.